FINAL-YEAR PROJECT | CIE-3

TEAM RSST

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ShieldWall

An eBPF/XDP engine that stops DDoS attacks on small servers in seconds and explains itself in plain English with gpt-4o-mini.

C | Python | eBPF/XDP

Detect, Mitigate, Explain

Small servers drown in fake traffic

37%

of all DDoS attacks hit the gaming industry, the most attacked sector

$38/hr

is all a DDoS-for-hire service costs. Attacking is cheap, defending is not

94%

of web DDoS attacks stay under 100k req/s, small enough to slip past basic detection

Our focus: small operators (game servers, small VPS owners, campus networks) who can't afford Cloudflare-scale protection and have nothing automatic today.

Sources: Akamai State of the Internet, SentinelOne, StationX DDoS Statistics 2026, Cloudflare

What the system has to do

1. **Detect fast.** Spot a flood within seconds from live traffic counts.
2. **Block automatically.** Drop attack traffic in the kernel with no human in the loop.
3. **Protect real users.** Allowlist, rule cap and expiring rules so players keep playing.
4. **Explain itself.** gpt-4o-mini writes plain-English summaries but never makes blocking decisions.
5. **Run on small hardware.** One commodity server, tested in an isolated lab.

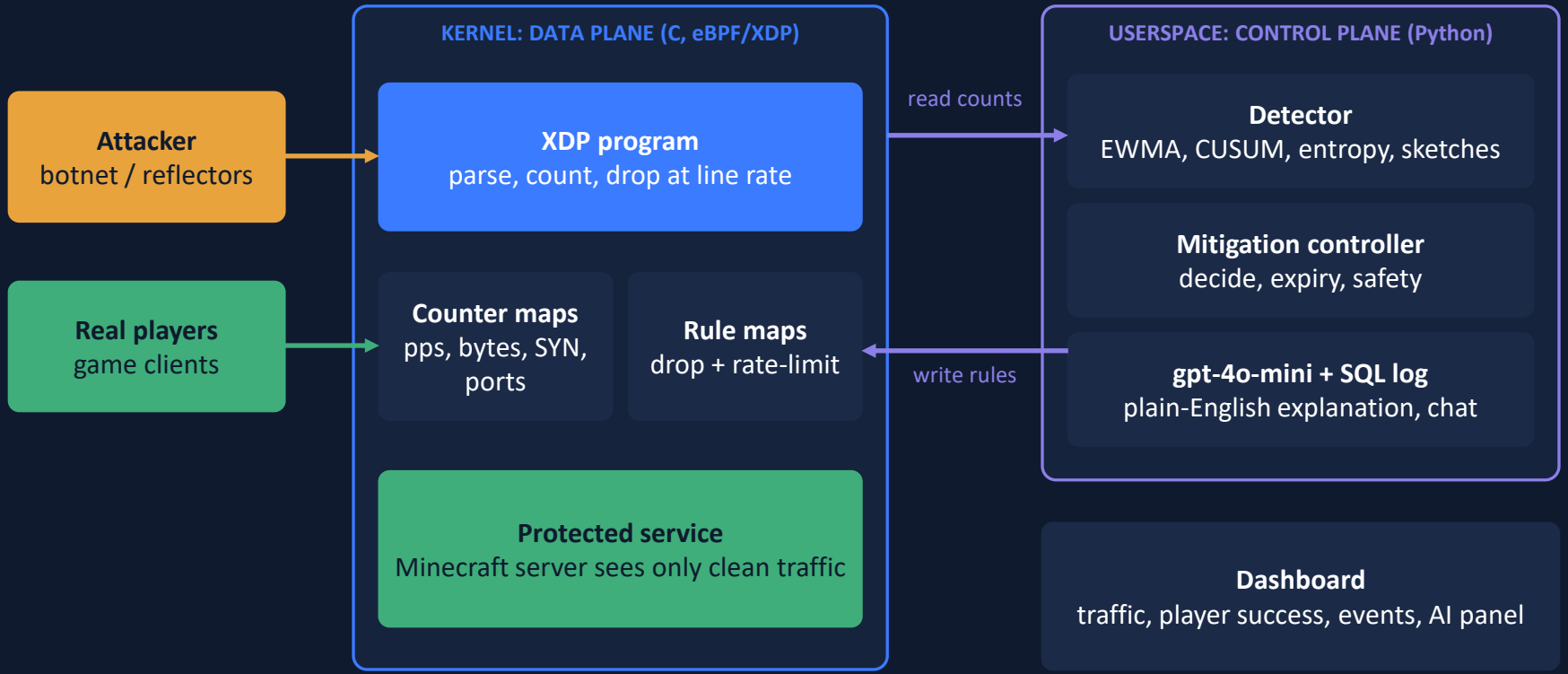
Known limit

Floods bigger than the internet link need upstream help, so ShieldWall asks the provider to filter them (BGP FlowSpec).

No training needed

It learns each server's normal traffic live. No dataset, no retraining.

Design: the kernel drops, userspace decides



Implementation: detect, mitigate, explain

Detect

Three methods vote

- EWMA baseline with z-score
- CUSUM catches slow ramps
- Entropy of source spread
- Count-Min, HyperLogLog and Bloom sketches track sources in KBs
- No training: learns normal traffic live

Mitigate

Block, then undo

- Signature drop (e.g. UDP src port 389)
- Token-bucket rate limit on unknown sources
- Every rule expires in ~60 s
- Allowlist, rule cap, dry-run mode

Explain

Off the hot path

- gpt-4o-mini writes the incident summary
- Chat over the SQL event log
- Checked: 0 made-up numbers
- Never makes a blocking decision

Stack: C + eBPF/XDP (kernel), Python + bpftool (control), SQLite, OpenAI gpt-4o-mini, scapy kernel test, live web dashboard, pytest + GitHub Actions

Evaluation and results

Four test scenarios

1. Clean attack
2. Attack mixed with real traffic
3. Legitimate spike, no attack
4. Attack bigger than the link

LLM check (gpt-4o-mini)

100% numbers grounded (49/49)
100% named the right attack
1.6 s reply, **$0.00008** each

Metric	Without	With ShieldWall
Time to detect (s)	No detection	**2 s**
Time to mitigate (s)	Never	**3 s**
Real traffic served in attack	30-50%	**98-99%**
False positives (scenario 3)	-	**0 false alarms**
Detection F1 (60 trials)	-	**F1 0.97**

Result: 51 of 52 attacks caught (median 2 s), 100% precision, zero false alarms. On a real Linux kernel the XDP program dropped 3,000 of 3,000 flood packets and passed every player packet.

Code and summary

GitHub repository

github.com/SushrithKbtech/shieldwall

XDP program, detection engine, dashboard, tests, kernel test and full results.

Real problem: Gaming is the top DDoS target and small operators are priced out of protection.

Real engineering: Kernel eBPF/XDP, streaming sketches, CUSUM detection and a grounded gpt-4o-mini explainer.

Real proof: 98-99% of player traffic served under attack, and 3,000 of 3,000 flood packets dropped on a real kernel.

A smart bouncer for small servers: it blocks attacks by itself and explains what it did.